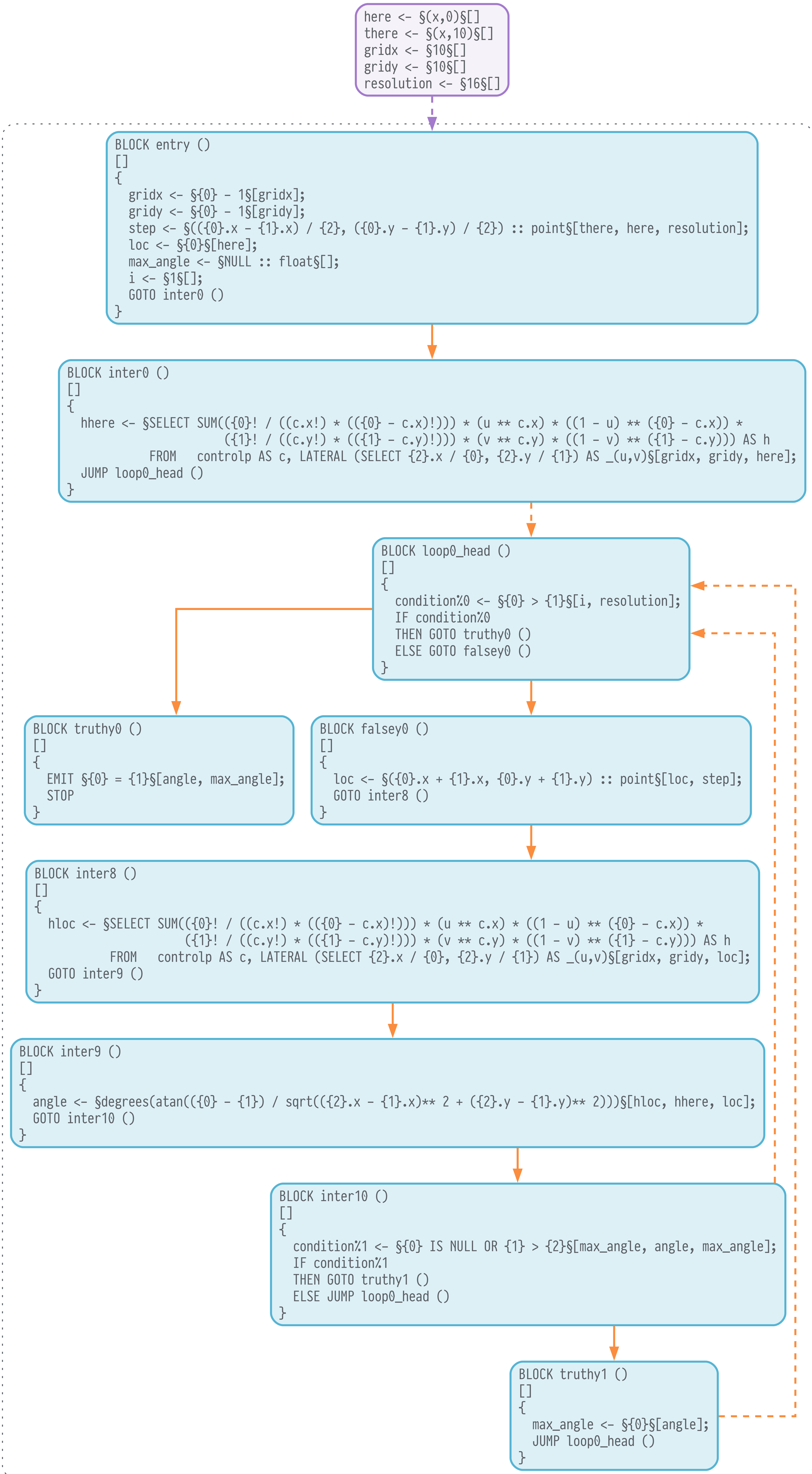
here <- §(x,0)§[]
there <- §(x,10)§[]
gridx <- §10§[]
gridy <- §10§[]
resolution <- §16§[]
BLOCK entry ()
[]
{
gridx <- §{0} - 1§[gridx];
gridy <- §{0} - 1§[gridy];
step <- §(({0}.x - {1}.x) / {2}, ({0}.y - {1}.y) / {2}) :: point§[there, here, resolution];
loc <- §{0}§[here];
max_angle <- §NULL :: float§[];
i <- §1§[];
GOTO inter0 ()
}
BLOCK inter0 ()
[]
{
hhere <- §SELECT SUM(({0}! / ((c.x!) * (({0} - c.x)!))) * (u ** c.x) * ((1 - u) ** ({0} - c.x)) *
({1}! / ((c.y!) * (({1} - c.y)!))) * (v ** c.y) * ((1 - v) ** ({1} - c.y))) AS h
FROM controlp AS c, LATERAL (SELECT {2}.x / {0}, {2}.y / {1}) AS _(u,v)§[gridx, gridy, here];
JUMP loop0_head ()
}
BLOCK loop0_head ()
[]
{
condition%0 <- §{0} > {1}§[i, resolution];
IF condition%0
THEN GOTO truthy0 ()
ELSE GOTO falsey0 ()
}
BLOCK truthy0 ()
[]
{
EMIT §{0} = {1}§[angle, max_angle];
STOP
}
BLOCK falsey0 ()
[]
{
loc <- §({0}.x + {1}.x, {0}.y + {1}.y) :: point§[loc, step];
GOTO inter8 ()
}
BLOCK inter8 ()
[]
{
hloc <- §SELECT SUM(({0}! / ((c.x!) * (({0} - c.x)!))) * (u ** c.x) * ((1 - u) ** ({0} - c.x)) *
({1}! / ((c.y!) * (({1} - c.y)!))) * (v ** c.y) * ((1 - v) ** ({1} - c.y))) AS h
FROM controlp AS c, LATERAL (SELECT {2}.x / {0}, {2}.y / {1}) AS _(u,v)§[gridx, gridy, loc];
GOTO inter9 ()
}
BLOCK inter9 ()
[]
{
angle <- §degrees(atan(({0} - {1}) / sqrt(({2}.x - {1}.x)** 2 + ({2}.y - {1}.y)** 2)))§[hloc, hhere, loc];
GOTO inter10 ()
}
BLOCK inter10 ()
[]
{
condition%1 <- §{0} IS NULL OR {1} > {2}§[max_angle, angle, max_angle];
IF condition%1
THEN GOTO truthy1 ()
ELSE JUMP loop0_head ()
}
BLOCK truthy1 ()
[]
{
max_angle <- §{0}§[angle];
JUMP loop0_head ()
}